

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)**

I B.Tech I Semester Regular Examinations, January-2024

ENGINEERING PHYSICS

(Common to CSE, CSE (AIML), CSE (AI) & CSE (DS))

Time: 3 hours

Max. Marks: 70

Note::

- i. Question 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
- ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Q. No.	Questions	BTL	CO	Marks
1.	a) State principal of superposition.	K1	CO1	2M
	b) Define dispersive power of a grating.	K1	CO1	2M
	c) Sketch the following planes of the cubic unit cell of (101) and (121).	K3	CO2	2M
	d) State Bragg's Law of X-ray diffraction.	K1	CO2	2M
	e) Define the terms (i) Dielectric polarizability (ii) Dielectric constant	K1	CO3	2M
	f) Why diamagnetic materials have negative susceptibility?	K1	CO3	2M
	g) Calculate the wavelength associated with an electron raised to potential of 54 V.	K3	CO4	2M
	h) Write any two draw backs of classical free electron theory.	K1	CO4	2M
	i) Define Energy band gap and write its physical significances.	K1	CO5	2M
	j) What is drift current?	K1	CO5	2M
UNIT-I				
2.	a) With the help of a neat diagram, describe the experimental arrangement to produce Newton's rings by reflected light. Prove that the diameter of dark ring is proportional to the square root of the natural number.	K2	CO1	8M
	b) In Newton's ring experiment, the diameters of 4th and 12th dark rings are 0.4 cm and 0.8 cm respectively. Find the diameter of 20th dark ring.	K4	CO1	2M
OR				
3.	a) Discuss in detail Fraunhofer diffraction due to double slit.	K2	CO1	7M
	b) Explain Diffraction Grating.	K2	CO1	3M
UNIT-II				
4.	a) Describe the seven crystal systems with neat diagrams.	K2	CO2	7M
	b) A substance with the FCC lattice has density 6200 kg/m ³ and molecular weight 60.2. Calculate the lattice constant a.	K4	CO2	3M
OR				
5.	a) Explain X-Ray Spectrometer.	K2	CO2	4M

	b)	Explain the determination of crystal structure by powder method.	K2	CO2	6M
UNIT-III					
6.	a)	What is electronic polarization? Show that the electronic polarization depends on the radius of the constituent atom.	K1	CO3	8M
	b)	If the radius of hydrogen atom is 0.055 nm, then calculate the electronic polarizability.	K3	CO3	2M
OR					
7.	a)	Define Hysteresis. Explain hysteresis loop observed in ferromagnetic materials.	K1	CO3	6M
	b)	Differentiate the types of magnets based on Hysteresis Loop.	K3	CO3	4M
UNIT-IV					
8.	a)	Obtain an expression for the energy of a particle enclosed in one-dimensional potential box of infinite height.	K5	CO4	7M
	b)	Find the energy of ground state electron moving in one dimension infinitely high potential box of width $10 \text{ \AA}$.	K4	CO4	3M
OR					
9.	a)	What is the difference between the classical and quantum free electron theory?	K1	CO4	4M
	b)	Derive expression for conductivity based on quantum free electron theory.	K3	CO4	6M
UNIT-V					
10.	a)	Deduce an expression for carrier concentration of electrons in the conduction band of an intrinsic semiconductor.	K3	CO5	7M
	b)	Explain the effective mass of electron.	K2	CO5	3M
OR					
11.	a)	State and explain Hall Effect. Derive an expression for Hall coefficient.	K1	CO5	8M
	b)	An n-type semiconductor specimen has a Hall coefficient (R_H) $3.66 \times 10^{-11} \text{ m}^3 \text{ C}^{-1}$. The conductivity of the specimen is found to be $112 \times 10^7 / \Omega \text{m}$. Calculate the charge carrier density (n).	K4	CO5	2M